

ORIGINAL RESEARCH

Hospitalizations and Mortality Among Medicare Beneficiaries at Cardiovascular Risk



Disruptions and Recovery From the COVID-19 Pandemic

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ABSTRACT

BACKGROUND The COVID-19 pandemic led to disruptions in cardiovascular care in mid-2020, but less is known about how patterns of care and clinical outcomes changed for older adults with cardiovascular (CV) risk factors and/or established cardiovascular disease (CVD) in the later stages of the pandemic.

OBJECTIVES This study sought to identify changes in utilization of health care services and all-cause mortality among Medicare beneficiaries with CV risk factors or CVD in the “late pandemic” (January 2021 to December 2022) overall and across subgroups based on rurality and social vulnerability index.

METHODS The study included all Medicare fee-for-service and Medicare Advantage beneficiaries with CV risk factors (hypertension, diabetes mellitus, hyperlipidemia) or CVD (coronary heart disease, heart failure, atrial fibrillation, stroke) from January 2018 to December 2022. A Poisson-lognormal regression model was fitted to compare the adjusted incidence rate ratio (aIRR) of acute care visits, outpatient visits, and all-cause mortality during the late pandemic (January 2021 to July 2022) compared with the prepandemic period (January 2018 to December 2019).

RESULTS There were 51,355,577 beneficiaries in the prepandemic period and 55,214,638 in the late pandemic period with CV risk factors or CVD. Hospital visit rates were lower in the late pandemic period compared with the prepandemic period (aIRR: 0.918; 95% CI: 0.915-0.922), and declines were evident among rural and urban beneficiaries, as well as across all levels of social vulnerability and in both fee-for-service and Medicare Advantage. In contrast, use of outpatient visits was higher during the late pandemic period (aIRR: 1.141; 95% CI: 1.134-1.149); these increases were on average higher among urban communities and Medicare Advantage beneficiaries. Mortality was higher during the late pandemic period (aIRR: 1.248; 95% CI: 1.240-1.257), and surges in mortality among this population tracked with national COVID-19 mortality. Beneficiaries in the most socially vulnerable communities (aIRR: 1.116; 95% CI: 1.106-1.124) and those covered by Medicare Advantage (aIRR: 1.342; 95% CI: 1.329-1.356) experienced the greatest increase.

CONCLUSIONS In the later stages of the COVID-19 pandemic, among Medicare beneficiaries with CV risk factors or established CVD, hospitalizations were lower than before the pandemic, outpatient care was used more frequently, and mortality was significantly elevated across risk groups and geographies, with the greatest increases seen in vulnerable communities and those covered by Medicare Advantage. (JACC. 2026;87:1313-1323) © 2026 by the American College of Cardiology Foundation.

ABBREVIATIONS AND ACRONYMS

CV = cardiovascular

CVD = cardiovascular disease

ED = emergency department

FFS = fee-for-service

MA = Medicare Advantage

The early phase of the COVID-19 pandemic led to morbidity and mortality as well as enormous disruptions in inpatient and outpatient care that were most pronounced among socially vulnerable populations. Older adults with cardiovascular (CV) risk factors or established cardiovascular disease (CVD) may

have been particularly susceptible both to COVID itself and to these care disruptions, as evidenced by precipitous declines in myocardial infarction and stroke hospitalizations,¹⁻⁴ and by associated increases in population-level CV mortality during the first few months of the pandemic.⁵⁻⁸ However, little is known about whether these patterns persisted into the later phases of the pandemic.

SEE PAGE 1324

Understanding changes in health care utilization and outcomes for older adults with CV risk factors or disease is critically important and could have major downstream implications for health care spending, payment policy, and population health. First, understanding the impact of COVID-19 on older adults with CVD could have implications for future COVID waves or indeed for future viral pathogens or pandemics more broadly. Second, understanding changes in care utilization is important. Disruptions in important health care services, including preventive screenings, treatment of risk factors such as diabetes and hypertension, and procedural care for CV conditions (eg, ischemic heart disease, stroke), could have led to a surge in pent-up hospitalizations after the early COVID period.^{9,10} On the other hand, patients could have continued to avoid seeking care as the pandemic went on because of fear of contracting the virus, been lost to follow-up, or experienced reduced access to care.¹¹⁻¹³ Strains on hospital capacity, increases in out-of-hospital deaths whether due to COVID or otherwise, or shifts in coding or

admission practices could have also affected patterns of care. In addition, understanding changes in health care services and clinical outcomes among socially vulnerable populations, who disproportionately experienced both the direct clinical impacts of COVID and health care disruptions,¹⁴⁻¹⁸ could help inform public health and policy strategies to improve CV health in the years that follow the pandemic.

Therefore, in this national study, we aimed to answer 3 questions. First, how did health care utilization among Medicare beneficiaries with CV risk factors or established CVD in the late pandemic period compare to pre-COVID levels? Second, how did mortality rates compare between the late COVID period and pre-COVID periods? And finally, how did any observed changes vary by rurality, social vulnerability, or fee-for-service (FFS) vs Medicare Advantage (MA) coverage?

METHODS

DATA SOURCES AND STUDY POPULATION. We used the Virtual Research Data Center (VRDC) to perform a longitudinal study of the Medicare population from January 1, 2018, through December 31, 2022. The VRDC captures 100% of beneficiary claims across all hospital-based (inpatient, observation, emergency department [ED]) and outpatient care settings, including both in-person and telehealth visits. We defined the pre-COVID period as January 1, 2018, through December 31, 2019, and the late COVID period as January 1, 2021, through December 31, 2022. We also obtained data on weekly national COVID-19 death counts per 100,000 population from the Centers for Disease Control and Prevention,¹⁹ and approximate start and end dates for major COVID variant waves from *The New York Times* COVID variant tracker.²⁰

The study population included all FFS and MA beneficiaries with CV risk factors (hypertension, diabetes mellitus, or hyperlipidemia) or established

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

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CVD (coronary heart disease, heart failure, atrial fibrillation, or stroke). Exclusions are shown in [Supplemental Figure 1](#). We followed Medicare's Chronic Conditions Warehouse (CCW) methodology to identify each qualifying condition. That algorithm uses diagnosis codes in claims during the preceding 12 to years to identify beneficiaries with these common clinical conditions, and is typically applied on an semiannual or annual basis.²¹ We adapted these algorithms to capture beneficiaries with CV risk factors or CVD on a rolling basis, thus reducing ascertainment bias associated with end-of-year comorbidity reporting and aligning at-risk periods more precisely by allowing entry into or exit from the cohort at any point in the study period. For our purposes, entrance into the surveillance cohort begins 1 day after the claim that qualifies a beneficiary as having a chronic condition (which would exclude the initial qualifying utilization event from our outcomes), and exit from the surveillance cohort occurs 1 day after the 1-year or 2-year period specified by CCW for each chronic condition, unless a beneficiary has an additional qualifying event that extends their membership in the at-risk cohort. We used the VRDC to determine characteristics of each beneficiary, including age, sex, race, and ethnicity, and dual enrollment in Medicaid. For race and ethnicity, we used the Research Triangle Institute variables, which augment enrollment data with additional information based on surname and residence.²² For subgroup analyses, in addition to FFS vs MA coverage, we assessed rurality and social vulnerability. We assessed rurality using Core-Based Statistical Areas based on county level Federal Information Processing Standard (FIPS) codes to determine whether beneficiaries lived in metropolitan or micropolitan areas, which are explicitly defined, or rural areas, which are undefined.²³ Beneficiaries' FIPS codes of residence were also used to determine the social vulnerability index (SVI) of their communities. The SVI was developed by the Centers for Disease Control and Prevention and combines U.S. Census Bureau data on 15 variables across 4 domains: socioeconomic status, housing composition and disability, minority status and language, and housing type and transport.^{10,24} The overall SVI provides a county-level percentile rank from lowest to highest vulnerability, with higher percentile ranks indicating higher levels of social vulnerability. We categorized beneficiaries into quartiles based on their national SVI rankings.

OUTCOMES. We evaluated hospital visits (inpatient, observation, and ED stays) and outpatient visits (in-person and telehealth) per beneficiary-year. In addition, we examined changes in all-cause mortality,

both overall and in-hospital (defined as death with an acute care encounter within ± 1 day).

STATISTICAL ANALYSIS. We first compared beneficiary sociodemographic characteristics during the pre- and late COVID periods. We then determined hospital visits, outpatient visits, and all-cause mortality, as well as incidence rates for each outcome per beneficiary-year, and plotted the raw data. To the overall mortality graph, we added a trace line representing the number of deaths ascribed to COVID per 100,000 population each month during the pandemic, as well as markers for each variant wave, as described above.¹⁹

A hierarchic bayesian Poisson-lognormal regression model was fitted to outcome counts during the late COVID (January 2021 to December 2022) compared with the pre-COVID period (January 2018 to December 2019),²⁵ adjusted for age, sex, race, and quarter-year with an offset term for beneficiary-years. We excluded the year 2020 from the model to focus on the long-term average impacts of the pandemic on the health care system rather than the acute disruptions, which have been documented in previous work;^{9,26} we also excluded at-risk time and outcomes for MA beneficiaries from July 2022 to December 2022 to accommodate delayed data delivery schedules for MA Encounter data where longer hospitalizations may affect our cohort ascertainment and outcomes quantification. Our model also included a state-level random effect as well as a variable indicating whether a beneficiary had a billing code indicating acute COVID-19 infection in the previous 90 days, ascertained similarly to our CCW definitions as described above. Full details of our modeling approach are provided in the [Supplemental Appendix](#). We then repeated our main analysis across 2 key subgroupings, rurality (metropolitan, micropolitan, rural), and social vulnerability (quartiles), and further presented results from our main analysis by enrollment in FFS vs MA. The FFS vs MA analysis was added post hoc and should be considered as exploratory.

In all models, we carefully specified relatively noninformative priors for all parameters, and we assessed posterior convergence diagnostic criteria, such as Rhat and traceplots for all parameters of interest within the VRDC environment, as required by our data use agreement. We observed no evidence of model fit, convergence, or posterior chain mixing issues. Full details of our modeling specification, prior parameterization, and explicit diagnostic assessment procedures with criteria are provided in the [Supplemental Appendix](#) for researchers interested in replicating our approach.

TABLE 1 Characteristics of Medicare Beneficiaries With Cardiovascular Risk Factors or Disease

	Prepandemic, Jan 2018 to Dec 2019 (n = 51,355,577)	Late Pandemic, Jan 2021 to Dec 2022 (n = 55,214,638)
Age at start month, y		
<65	13.5	11.8
65-74	49.5	51.2
75-84	25.9	26.6
≥85	11.1	10.4
Sex		
Male	44.4	44.5
Female	55.6	55.5
Race or ethnicity ^a		
White	75.0	73.9
Black	10.8	10.8
Hispanic	8.1	8.5
Other	6.1	6.7
Medicare enrollment type		
Fee-for-service	62.4	53.3
Medicare Advantage	37.6	46.7
Rurality ^b		
Urban	82.1	82.5
Micropolitan	10.1	10.0
Rural	7.8	7.6
Social Vulnerability Index ^c		
Quartile 1 (least vulnerable)	13.9	14.2
Quartile 2	24.6	24.7
Quartile 3	29.2	29.1
Quartile 4 (most vulnerable)	32.3	32.0
Chronic conditions		
Acute myocardial infarction	3.3	3.4
Atrial fibrillation	17.5	17.7
Diabetes	37.4	36.9
Hyperlipidemia	82.5	84.8
Hypertension	83.5	82.6
Ischemic heart disease	29.3	29.1
Stroke/transient ischemic attack	13.8	12.8

Percentages of beneficiary-months are presented unless otherwise specified. For chronic conditions, only cardiovascular risk factors or cardiovascular diseases are presented. ^aRace/ethnicity was determined based on Research Triangle Institute categories. ^bRurality was defined based on Core-Based Statistical Areas. ^cSocial Vulnerability Index of beneficiaries' place of residence was categorized in national quartiles.

This study was approved by the Office of Human Research Protection at Washington University in St Louis, and the requirement for informed consent was waived owing to the de-identified nature of the data. Analyses were performed using SAS Enterprise Guide 8.3, R version 4.4.1,²⁷ and Stan version 2.32.2.²⁸ *P* values <0.001 were considered to be statistically significant.

RESULTS

There were 51,355,577 and 55,214,638 unique Medicare beneficiaries with CV risk factors or disease in in

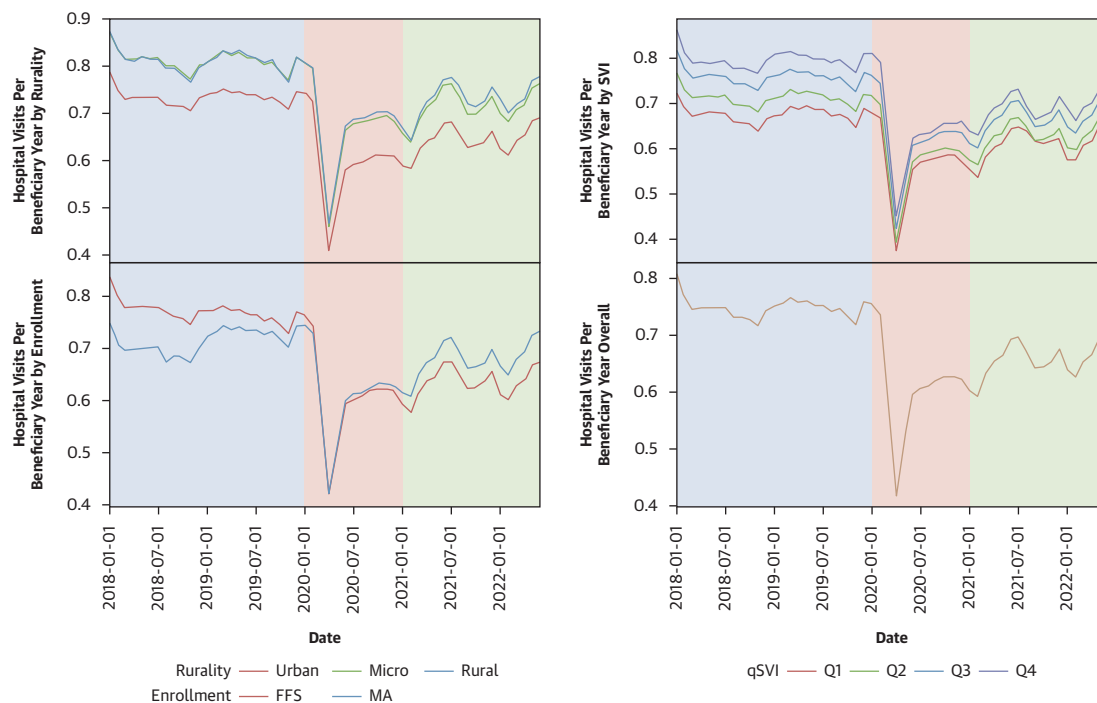
the 2018-2019 and 2021-2022 time periods, respectively, constituting a total of 206,781,871 beneficiary-years of data (Table 1). Beneficiary characteristics were generally similar between the pre- and late pandemic periods, although in the later period beneficiaries were slightly older, less often White (75.0% vs 73.9%), more often urban (82.1% vs 82.5%), and more likely to be enrolled in MA (37.6% vs 46.7%).

HOSPITAL VISITS. Unadjusted trends in hospital visits per beneficiary-year are shown in Figure 1, both overall and across key subgroups. Hospital visit rates were higher among rural beneficiaries compared with their urban counterparts over the entire study period. In addition, beneficiaries residing in communities with high social vulnerability (quartile 4) had markedly higher hospital visit rates than those in the lowest (quartile 1). Rural beneficiaries had higher hospital visit rates than their urban counterparts. Patterns were similar when these visits were disaggregated into ED only, observation, and inpatient stays (Supplemental Figures 2 to 4).

Model results suggested that, on average, hospital visit rates were statistically significantly lower in the late COVID period compared with the pre-COVID period (adjusted incidence rate ratio [aIRR]: 0.918; 95% CI: 0.915-0.922) (Table 2). Declines in inpatient care were similar across urban (aIRR: 0.895; 95% CI: 0.890-0.900), micropolitan (aIRR: 0.873; 95% CI: 0.867-0.884), and rural (aIRR: 0.877; 95% CI: 0.870-0.884) areas. These patterns were also evident across levels of SVI, with a lower drop among the least vulnerable (aIRR: 0.895; 95% CI: 0.889-0.901) than among the most vulnerable (aIRR: 0.875; 95% CI: 0.870-0.880; *P* for interaction <0.001) beneficiaries. FFS beneficiaries experienced steeper declines in hospital visits on average (aIRR: 0.885; 95% CI: 0.881-0.890) than those enrolled in MA (aIRR: 0.967; 95% CI: 0.962-0.973; *P* for interaction <0.001). Findings were similar when these visits were disaggregated into ED only, observation, and inpatient stays (Supplemental Table 1).

OUTPATIENT VISITS. Unadjusted trends in outpatient visit rates (in-person and telehealth combined) are shown in Figure 2. Differences between subgroups were generally similar to those for hospital visits, with the exception of rural beneficiaries, who had lower outpatient visit rates (but higher hospital visit rates) than their urban counterparts. When these visits were disaggregated into in-person and telehealth visits, it was evident that the majority of the growth in outpatient visits was due to an increase in telehealth (Supplemental Figures 5 and 6)

FIGURE 1 Hospital Visits Among Medicare Beneficiaries With Cardiovascular Risk Factors or Disease, January 2018 to June 2022



Observed trends in the number of hospital visits per 30 beneficiary-days, defined as an inpatient, observation, or emergency department visit, among Medicare beneficiaries with cardiovascular risk factors or established cardiovascular disease from January 2018 through July 2022, overall (bottom right) and by rurality (top left), Social Vulnerability Index quartile (qSVI; top right), and fee-for-service (FFS) vs Medicare Advantage (MA) enrollment (bottom left). Shading reflects our study's prepandemic (2018-2019), early COVID (2020), and late pandemic (2021-2022) periods.

Model results suggested that outpatient visits (combined in-person and telehealth) were higher during the late COVID period than in the pre-COVID period (aIRR: 1.141; 95% CI: 1.134-1.149) (Table 2). Increases in outpatient visits were greater among beneficiaries in urban areas (aIRR: 1.090; 95% CI: 1.084-1.096) than among those residing in micropolitan (aIRR: 1.025; 95% CI: 1.018-1.031; P for interaction <0.001) or rural (aIRR: 1.004; 95% CI: 0.997-1.011; P for interaction <0.001) areas. Increases were qualitatively similar among all levels of SVI (aIRRs: 1.054, 1.053, 1.049, and 1.057 for least to most vulnerable). The rise in outpatient visits was more pronounced among MA beneficiaries (aIRR: 1.215; 95% CI: 1.204-1.225) than those in FFS (aIRR: 1.093; 95% CI: 1.084-1.102; P for interaction <0.001).

ALL-CAUSE MORTALITY. Unadjusted trends in mortality are shown in Figure 3, and were higher among rural compared with urban beneficiaries, and among the most vs least socially vulnerable beneficiaries. Peaks and troughs in mortality tracked largely with national COVID mortality. Model results suggested

that mortality was higher during the late COVID period than in the pre-COVID period (aIRR: 1.248; 95% CI: 1.240-1.257) (Table 2). Increases in mortality were similar in magnitude for urban (aIRR: 1.076; 95% CI: 1.069-1.083), micropolitan (aIRR: 1.078; 95% CI: 1.068-1.089) and rural (aIRR: 1.089; 95% CI: 1.077-1.100) beneficiaries. Increases in mortality were larger in the more socially vulnerable areas (aIRR: 1.035, 1.056, 1.080, and 1.116 for SVI quartiles from least to most vulnerable; $P < 0.001$ for quartiles 3 and 4 vs 1). Increases in all-cause mortality were higher in Medicare Advantage (aIRR: 1.342; 95% CI: 1.329-1.356) than fee-for-service (aIRRs: 1.187; 95% CI: 1.176-1.197; P for interaction <0.001). When we analyzed the subset of only in-hospital mortality, the increases were slightly smaller (aIRR: 1.136; 95% CI: 1.127-1.145) overall and among key subgroups (Supplemental Table 2), although the overall patterns were similar, with no difference in the increase in mortality by rurality, a higher increase among more vulnerable areas, and a higher increase in MA.

TABLE 2 Health Care Utilization Among Medicare Beneficiaries With Cardiovascular Risk Factors or Disease

	Incidence Rate Ratio (95% CI), Late vs Prepandemic	P Value for Prepandemic vs Late Pandemic	P Value for Interaction
Hospital visits			
Overall	0.918 (0.915-0.922)	<0.001	
Rurality			
Urban	0.895 (0.890-0.900)	<0.001	Ref
Micropolitan	0.873 (0.867-0.879)	<0.001	0.92
Rural	0.877 (0.870-0.884)	<0.001	0.035
SVI			
Quartile 1 (least vulnerable)	0.895 (0.889-0.901)	<0.001	
Quartile 2	0.888 (0.883-0.893)	<0.001	0.56
Quartile 3	0.879 (0.874-0.884)	<0.001	<0.001
Quartile 4 (most vulnerable)	0.875 (0.870-0.880)	<0.001	<0.001
Medicare			
Fee-for-service	0.885 (0.881-0.890)	<0.001	
Medicare Advantage	0.967 (0.962-0.973)	<0.001	<0.001
Outpatient visits			
Overall	1.141 (1.134-1.149)	<0.001	
Rurality			
Urban	1.090 (1.084-1.096)	<0.001	
Micropolitan	1.025 (1.018-1.031)	<0.001	<0.001
Rural	1.004 (0.997-1.011)	0.286	<0.001
SVI			
Quartile 1 (least vulnerable)	1.054 (1.047-1.060)	<0.001	
Quartile 2	1.053 (1.046-1.059)	<0.001	0.042
Quartile 3	1.049 (1.043-1.055)	<0.001	<0.001
Quartile 4 (most vulnerable)	1.057 (1.050-1.063)	<0.001	<0.001
Medicare			
Fee-for-service	1.093 (1.084-1.102)	<0.001	
Medicare Advantage	1.215 (1.204-1.225)	<0.001	<0.001
Mortality			
Overall	1.248 (1.240-1.257)	<0.001	
Rurality			
Urban	1.076 (1.069-1.083)	<0.001	
Micropolitan	1.078 (1.068-1.089)	<0.001	0.083
Rural	1.089 (1.077-1.100)	<0.001	0.031
SVI			
Quartile 1 (least vulnerable)	1.035 (1.024-1.046)	<0.001	
Quartile 2	1.056 (1.048-1.064)	<0.001	0.0017
Quartile 3	1.080 (1.072-1.088)	<0.001	<0.001
Quartile 4 (most vulnerable)	1.116 (1.106-1.124)	<0.001	<0.001
Medicare			
Fee-for-service	1.187 (1.176-1.197)	<0.001	
Medicare Advantage	1.342 (1.329-1.356)	<0.001	<0.001

SVI = social vulnerability index.

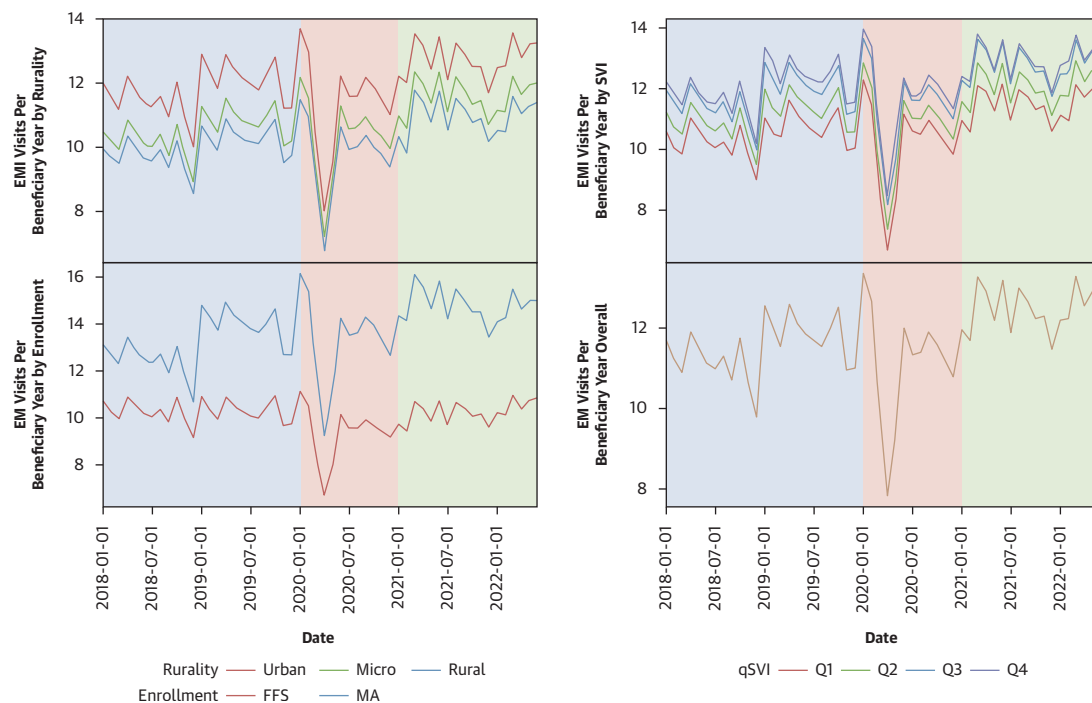
DISCUSSION

In this national study of all Medicare beneficiaries with documented CV risk factors or established CVD, we found that, on average, hospital visits were lower in 2021 and 2022 than before COVID-19. In contrast, outpatient visit rates increased beyond levels seen

before the pandemic, driven largely by a rise in telehealth visits. Mortality rates were significantly higher among beneficiaries with CV risk factors or conditions in 2021 and 2022 than before COVID.

Our first finding was an increase in outpatient visits, particularly telehealth, and a decrease in hospitalizations. Although this observational study does

FIGURE 2 Outpatient Visits Among Medicare Beneficiaries With Cardiovascular Risk Factors or Disease, January 2018 to June 2022



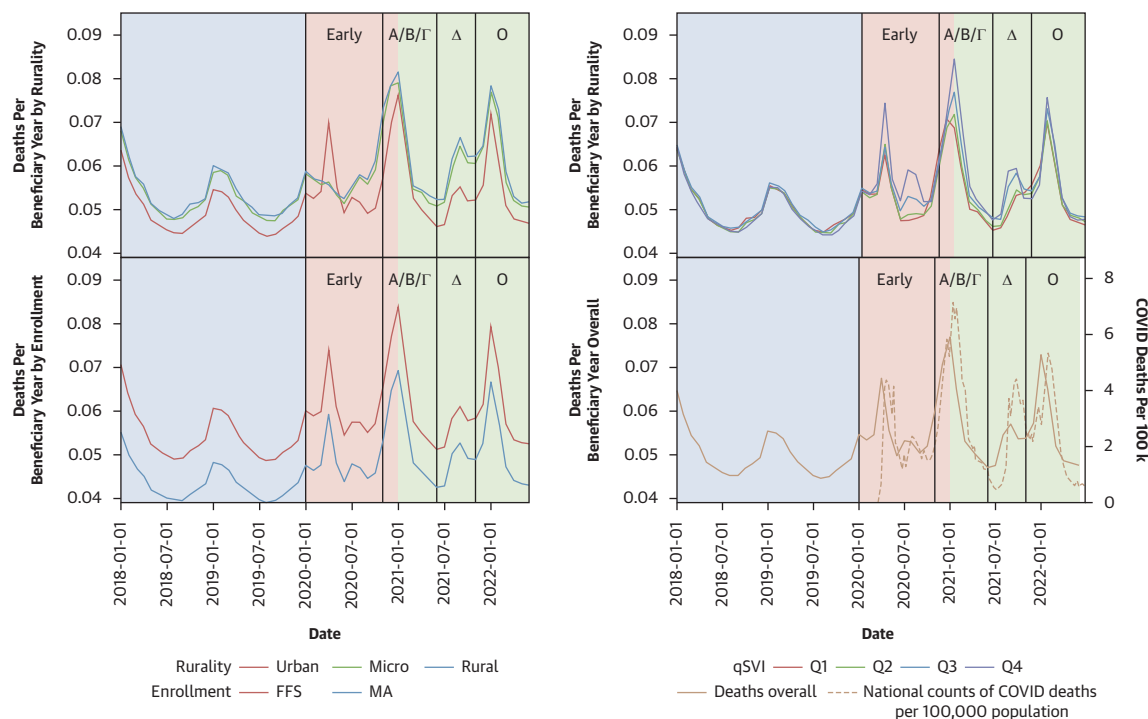
Observed trends in the number of outpatient visits per 30 beneficiary-days, defined as an in-person or telehealth evaluation and management visit, among Medicare beneficiaries with cardiovascular risk factors or established cardiovascular disease from January 2018 through July 2022, overall (bottom right) and by rurality (top left), qSVI (top right), and FFS vs MA enrollment (bottom left). Shading reflects our study's prepandemic (2018-2019), early COVID (2020), and late pandemic (2021-2022) periods. Abbreviations as in [Figure 1](#).

not allow causal inference, there are a number of potentially contributing factors to the patterns we observed. One possibility is that the pandemic drove durable changes in how people utilize health care; the increase in telehealth has been broadly documented,²⁹ and though debate remains about whether telehealth care is a substitute for in-person care or an add-on, it is likely here to stay. Indeed, greater use of outpatient care, which has traditionally been considered to be underutilized—particularly for high risk groups—and a higher threshold to seek hospital care are often seen as optimal use patterns. Lower use of hospital care could also be due to a healthy survivor bias, if the post-COVID-wave survivor cohorts were healthier and thus demonstrated less care-seeking behavior.

However, the fact that we found these patterns paradoxically coupled with higher mortality raises concerns, but also has several potential explanations. Some of the increased mortality was certainly due to

acute COVID infection; although we controlled for the presence of COVID diagnosis codes in our models, the waves of mortality evident in the raw mortality graphs emphasize that COVID, much like flu before it, likely drove seasonal variations in disease exacerbation and mortality among patients with CV risk factors or disease.³⁰ Indeed, peaks in mortality closely mirrored the timing of major COVID-19 waves—including Spring 2020, Winter 2020-2021, Fall 2021 (Delta), and Winter 2021-2022 (Omicron)—suggesting that the observed increases in mortality during the late period were largely driven by these infection surges. Some of the elevated mortality may also be driven by other after-effects of the virus rather than its acute impact. For example, COVID-19 leads to higher levels of inflammation, which contributes to worse overall clinical CV outcomes over both short long terms.³¹⁻³³ In addition, social and behavior risk factors (eg, poverty, dietary patterns, psychosocial stress) that are strongly linked with cardiometabolic

FIGURE 3 Mortality Among Medicare Beneficiaries With Cardiovascular Risk Factors or Disease, January 2018 to June 2022



Observed trends in mortality per 30 beneficiary-days, among Medicare beneficiaries with cardiovascular risk factors or established cardiovascular disease from January 2018 through July 2022, overall (bottom right; the dotted line represents national counts of COVID deaths as reported by the Centers for Disease Control and Prevention, and Greek characters represent different dominant variants of COVID-19) and by rurality (top left), qSVI (top right), and FFS vs MA enrollment (bottom left). Shading reflects our study's prepandemic (2018-2019), early COVID (2020), and late pandemic (2021-2022) periods. Abbreviations as in [Figure 1](#).

health worsened during the initial phase of the pandemic.³⁴⁻³⁸

System-level effects could also contribute, although these are difficult to study given data limitations. Hospital capacity challenges, including staff shortages, care delivery challenges, or constrained resources, could have affected outcomes in the acute or chronic setting, for both COVID and non-COVID illnesses, particularly during major spikes in COVID activity. Changes in coding or admission practices could confound our findings. Patient factors could also contribute; previous studies showed that people deferred care for heart attacks and strokes during the early days of the pandemic and often arrived at the hospital with more serious and complicated versions of their clinical syndrome than would otherwise have occurred.^{12,39,40} An increase in out-of-hospital deaths could also contribute; the increase we found in all-cause mortality was greater than the increase we found in in-hospital mortality, though such evidence is circumstantial at best. Together, both direct and

indirect effects of the pandemic may have contributed to the persistently higher mortality rates that we observed through the end of 2022.

The differences we found by geography and by social risk group also are important. For example, urban beneficiaries experienced a significant rise in outpatient visits driven by telehealth use, whereas rural beneficiaries—who already faced barriers for access to outpatient CV care⁴¹—experienced no change. People living in the most socially vulnerable counties experienced the largest increase in all-cause mortality, and the widening of disparities in the late pandemic period suggests that there is great need for improvements in health care access and infrastructure in vulnerable communities. These communities, which were hardest hit in the early waves of COVID in terms of both health and economic strain,⁴² may have lost key health infrastructure or access points that enabled use of nonemergency health care services for COVID or other illnesses. Aid given to businesses and health care organizations during the

pandemic often disproportionately benefited those with the infrastructure to easily claim it, potentially worsening structural and systemic inequities in our health care system.⁴³ Overall, our findings support the urgent need for targeted health system, community, and public health to prevent geographic disparities in outpatient care and CV health from widening further.

Finally, we also observed major differences across Medicare plan type. MA beneficiaries experienced less decline in hospital visits, but a greater increase in outpatient visits, as well as a greater increase in mortality, than those enrolled in traditional FFS Medicare. During the study period, beneficiaries were steadily shifting into MA, and the proportion of beneficiaries switching back and forth between MA and FFS was also growing;⁴⁴ although our analyses controlled for basic demographics and comorbidities, as well as area of residence, it is possible that these patterns reflect residual confounding related to the population joining MA. Given that this analysis was post hoc, it should be considered as exploratory, and future studies should evaluate these changes with more clinically rich data.

Our findings build on a recent analysis that found that ED visits were lower but rates of admission from the ED higher for non-COVID-related conditions after the onset of the pandemic.⁴⁵ We evaluated a comprehensive measure of overall acute care utilization (inpatient stays, ED visits, observation stays) in a higher risk population—those with CV risk factors or disease—who could potentially be more susceptible to the effects of the pandemic. In addition, our study provides new insights on changes in outpatient visits as well as a concerning sustained increase in all-cause mortality among people with CV risk factors or disease through 2022. Together, these findings build on early studies highlighting the direct and indirect tolls of the pandemic^{8,37,46–51} by providing a comprehensive evaluation of longer-term changes in health care utilization and mortality in the entire Medicare population, both overall and by geographic and social vulnerability.

STUDY LIMITATIONS. First, Medicare claims data do not include granular clinical information, and as such we were unable to assess patients' blood pressure, diabetes control, or other biometric or laboratory values. Second, we studied patients with CV risk factors or overt CVD in order to focus on a population with defined care needs, but our findings may not apply to people without these conditions, and we did

not explicitly account for clinical or claims-based indicators, such as medication adherence patterns, non-CV chronic conditions, or complex utilization measures such as frailty in our analysis, which may serve to further explain changes in utilization and mortality patterns from pre- to postpandemic. Third, associations seen in observational data do not imply causality, and further work is needed to disentangle the multiple potential mechanisms we outlined above. Our analytic choices also matter; this analysis compared marginal averages before and during the late pandemic period, which emphasizes persistent differences in care and outcomes, and allows the flexibility to investigate differential changes in levels by subgroupings of interest, such as rurality, enrollment choice, and measures describing areal social vulnerability. Use of an interrupted time series design may have revealed additional collapse-and-rebound dynamics at the cost of flexibility to investigate multiple differential changes by subgroup. A strength of this analysis is that it included all Medicare beneficiaries with CV risk factors or disease, including those enrolled in FFS or MA; this allowed us essentially complete capture of hard endpoints such as hospitalization and mortality for the older adult U.S. population across a 5-year period.

CONCLUSIONS

We found that for patients with CV risk factors or established CVD, hospitalizations remained lower in the late COVID-19 period than before COVID-19, outpatient care was used more frequently, and, most importantly, mortality remained significantly elevated across risk groups and geographies, with the greatest increases seen in the most vulnerable communities.

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APPENDIX For supplemental material, please see the online version of this paper.